

MAT1002 Lecture 20,

Outline

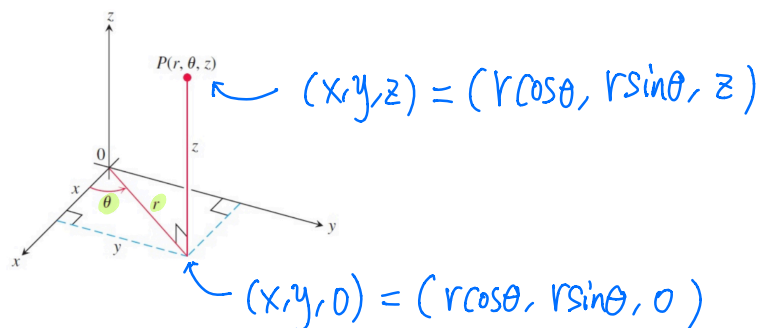
- Cylindrical coordinates
- Change of variable formula

Cylindrical Coordinates

Given any point P in the xyz -space, we can represent the point by (r, θ, z) , where

- ▶ r and θ are polar coordinates of the projection of P onto the xy -plane, and;
- ▶ z is the z -coordinate of P in the Cartesian coordinates.

The coordinate system above in (r, θ, z) is called the **cylindrical coordinate system**.



Cylindrical coordinates may be helpful in computing triple integrals when the solid E of integration has the form

$$E = \{(x, y, z) : (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\},$$

where D (the projection of E onto the xy -plane) is convenient to describe in polar coordinates. If D is given by

$$\alpha \leq \theta \leq \beta, \quad 0 \leq h_1(\theta) \leq r \leq h_2(\theta),$$

$(\beta - \alpha \leq 2\pi)$

then

$$\begin{aligned} \iiint_E f(x, y, z) dV &= \iint_D \left(\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz \right) dA \\ &= \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(x, y)}^{u_2(x, y)} f(r \cos \theta, r \sin \theta, z) dz \, r \, dr \, d\theta \end{aligned}$$

e.g. A solid E lies inside the cylinder $x^2 + y^2 = 1$, below the plane $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$. The density at a point $P(x, y)$ is $K\sqrt{x^2 + y^2}$ for some constant K . Find $\text{mass}(E)$.

Ans $\frac{12\pi}{5}K$.

Exercise

Use cylindrical coordinates to evaluate the integral

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2 + y^2) dz dy dx.$$

Ans $16\pi/5$.

Moral of the story: $dV = r dz dr d\theta$ after changing from xyz - to cylindrical coordinates.

Change of Variable Formula (Double Integrals)

When discussing double integrals in polar coordinates last week, we saw that

$$\iint_D f(x, y) dA = \iint_R f(r \cos \theta, r \sin \theta) r dr d\theta,$$

where R is the polar region corresponding to the region D (which is in the xy -plane). Essentially, we made the substitution

$$x = r \cos \theta, \quad y = r \sin \theta,$$

and replaced the differential dA with $r dr d\theta$. More generally, if we make the substitution

$$x = g(u, v), \quad y = h(u, v),$$

what would the formula become?

We first recall a fact about determinants: if $\vec{a} = \langle a, b \rangle$ and $\vec{b} = \langle c, d \rangle$, then area of $\triangle \vec{a} \vec{b} = \left| \det \begin{bmatrix} a & c \\ b & d \end{bmatrix} \right|$.

Another fact about determinants is that you can "take out" any constant from a row or from a column:

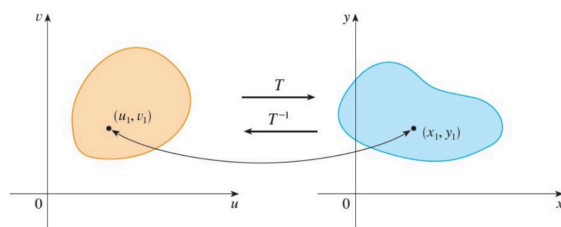
$$\det \begin{bmatrix} \lambda a & \lambda c \\ \lambda b & \lambda d \end{bmatrix} = \lambda \det \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \lambda^2 \det \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Suppose that

$$x = g(u, v) \quad \text{and} \quad y = h(u, v),$$

where g and h have continuous partial derivatives. We may understand this as a transformation T that maps a point in the uv -plane to a point in the xy -plane:

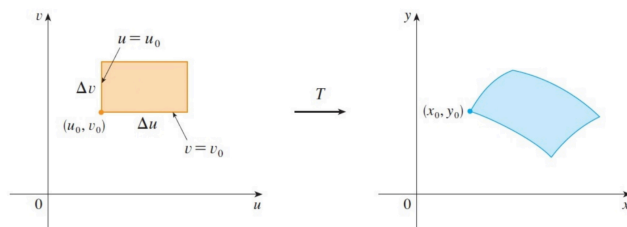
$$T(u, v) := (g(u, v), h(u, v)) = (x, y).$$



Let R be a region in the uv -plane, and let D be the image of R under the transformation T . That is,

$$D = T(R) := \{T(u, v) : (u, v) \in R\}.$$

Partition R into rectangles R_{ij} , and let D_{ij} be the image of R_{ij} under T .



Let $A(D_{ij})$ and $A(R_{ij})$ be the area of D_{ij} and R_{ij} , respectively.
When the norm of the partition is small,

$$A(D_{ij}) \approx \left| \frac{\partial(x, y)}{\partial(u, v)} \right| A(R_{ij}) = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \Delta u_i \Delta v_j, \quad \text{absolute value of the Jacobian}$$

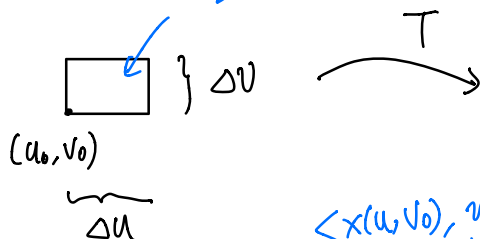
where
Jacobian, which is a det

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \leftarrow \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}$$

The determinant $\frac{\partial(x, y)}{\partial(u, v)}$ above is called the **Jacobian** of the transformation.

Details:

Some R_{ij}



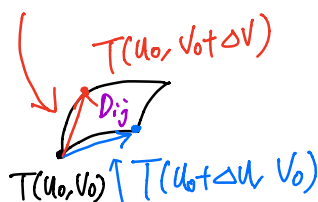
$$\langle x(u, v_0), y(u, v_0) \rangle$$

Fix v_0 , let $\vec{r}_1(u) := \underline{T(u, v_0)}$

Fix u_0 , let $\vec{r}_2(v) := \underline{T(u_0, v)}$

$$\langle x(u_0, v), y(u_0, v) \rangle$$

$$\vec{r} = T(u_0, v_0 + \Delta v) - T(u_0, v_0) = \vec{r}_2(v_0 + \Delta v) - \vec{r}_2(v_0) \approx \vec{r}_2'(v_0) \Delta v$$



$$\vec{r} = T(u_0 + \Delta u, v_0) - T(u_0, v_0)$$

$$= \vec{r}_1(u_0 + \Delta u) - \vec{r}_1(u_0)$$

$$\approx \vec{r}_1'(u_0) \Delta u$$

Hence $A(D_{ij}) \approx \text{Area}(\vec{r}, \vec{r}) = \left| \det [\vec{r}_1 \vec{r}_2] \right|$

$$= \left| \det \begin{bmatrix} \frac{\partial x}{\partial u} \Delta u & \frac{\partial x}{\partial v} \Delta v \\ \frac{\partial y}{\partial u} \Delta u & \frac{\partial y}{\partial v} \Delta v \end{bmatrix} \right| = \Delta u \Delta v \left| \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} \right|$$

Evaluated at (u_0, v_0)

$\frac{\partial(x, y)}{\partial(u, v)}$ "Jacobian of T "

In terms of differentials, one can memorize the above as

$$dA = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

Theorem (Change of Variables in Double Integrals)

Let f be continuous on a region D in the xy -plane. Suppose that

- ▶ the transformation T given by $x = g(u, v)$ and $y = h(u, v)$ maps a region R in the uv -plane onto D , and;
- ▶ the transformation is one-to-one in the interior of R , and;
- ▶ all partial derivatives of g and h are continuous.

Then

$$\iint_D f(x, y) dA = \iint_R f(g(u, v), h(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

e.g. (Polar transformation)

If a change of variables is given by the transformation

$$x = g(r, \theta) = r \cos \theta \quad \text{and} \quad y = h(r, \theta) = r \sin \theta,$$

then it follows that

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \geq 0,$$

which gives

$$\iint_D f(x, y) dA = \iint_R f(r \cos \theta, r \sin \theta) r dr d\theta.$$

This is consistent with the formula we stated in ~~Week 9~~.

last week

To satisfy the one-to-one condition, here we do not allow $r < 0$, i.e., domain of T is $\{(r, \theta) : r \geq 0, \theta \text{ is in an interval with length} = 2\pi\}$

Example

Evaluate the following integrals.

- (a) $\iint_D e^{(x+y)/(x-y)} dA$, where D is the trapezoidal region with vertices $(1, 0)$, $(2, 0)$, $(0, -2)$ and $(0, -1)$.

- (b) $\int_1^2 \int_{1/y}^y \sqrt{\frac{y}{x}} e^{\sqrt{xy}} dx dy$.

Ans

(a) $\frac{3}{4}(e - e^{-1})$

(b) $2e(e-2)$

Change of Variable Formula (Triple Integrals)

There is a similar change of variables formula for triple integrals.

Let T be a transformation given by

$$x = g(u, v, w), \quad y = h(u, v, w) \quad \text{and} \quad z = k(u, v, w),$$

which maps a region R in the uvw -space onto a region E in the xyz -space. The **Jacobian** of T is the following 3×3 determinant:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} := \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}.$$

Under assumptions similar to those in the theorem of change of variables for double integrals, the following formula holds for triple integrals:

$$\begin{aligned} & \iiint_E f(x, y, z) dV \\ &= \iiint_R f(g(u, v, w), h(u, v, w), k(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw. \end{aligned}$$